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Experience

Major League Baseball (MLB), *New York, NY*

Senior Software Engineer, Streaming Video Platform *March 2023 – Present*

- Architect and produce content discovery and delivery system for MLB's flagship streaming product, "MLB.tv", as well as internal Media Asset Managment (MAM) system for content manipulation and distribution. The distributed system is responsible for delivering live & on-demand baseball content for internal operations and external users. At peak, this system has delivered content to over 500,000 concurrent users with a p99 response time of 58 milliseconds.
- Technical Product Owner and Lead Engineer for a suite of tools which allows Tier-0 technical support to perform incident resolution on backend systems. Also provides a cross-cutting interface for event auditing, health monitoring, disaster recovery, and lower environment testing. Reduced direct on-call Engineer intervention by 75% within 1 month of adoption, and 97% within 6 months.
- Designed a server-driven client caching framework using GraphQL Subscriptions, allowing Frontend systems to display real-time data changes to users. Reduced programmatic polling to sensitive backend infrastructure by 99.99%, enabling system optimizations which allowed for 10% faster processing speed on business-critical operations.
- Extended an existing event-driven distributed system to handle new Product domains, representing 300% more traffic and video content for a given day. Proposed & implemented infrastructure extensions to persistence layers and event-driven autoscaling (KEDA) in anticipation of elevated load. The upgraded system handles 165 peak writes per second to Elasticsearch, with a p99 of 15 seconds end-to-end processing time for video availability.
- Identified and resolved recurring service outages in critical content discovery backend, eliminating 40 minutes of downtime every week during off-hours. Restored data store to five-nines uptime for service ecosystem which supports peak traffic of 800k requests per hour.
- Leveraged AI-assisted development to prototype and refactor automated video production workflows for HLS (HTTP Live Streaming) packaging, from initial POC scaffolding through modernization of legacy encoding scripts. Established architecture enabling dynamic resolution and multi-audio track management per HLS specification.
- Led cross-team initiative to enable 1080p video editing for Editorial teams, replacing legacy 720p workflow. Rapid-prototyped POC, authored RFC, and coordinated API contracts across frontend, middleware, and platform teams. Enabled multilingual audio track selection across 324 annual broadcasts for web and social media publishing, unlocking 648 additional video/audio combinations, representing a 3x content expansion.
- Led and executed on a business initiative to enable playback of 27,000 unique video and radio broadcasts for historical games on-demand. Wrangled legacy broadcast information and sanitized data to enable playback via the MLB.tv product, unlocking access to previously-unavailable video content for users.
- Evangelized software testing strategies which were applied across multiple backend teams. Logged Architecture Decision Records (ADRs) and developed new testing patterns which allowed for more expressive tests as documentation, improving code coverage by 20%. Correlated with reduced change failure rate of 5%.
- Designed and implemented a support ticketing system to streamline application support, routing user requests to a tracked log with automated team tagging, SLA capture, and AI-powered conversation summarization. Achieved 100% reduction in on-call Engineers as front-line support, reclaiming ~2-3 weekly off-hours interruptions. Established a foundation for future agentic automation including historical analysis, issue triage, and automated response to common support patterns.
- Championed and led design and implementation of a dedicated configuration management service. Authored the ADR, designed the domain architecture, and delivered implementation with a junior engineer. Achieved near-100% reduction in Engineering hours dedicated to configuration management, reclaiming approximately 4 engineer-weeks annually while shifting ownership to specialized Operators.

Capsule Pharmacy, *New York, NY*

Senior Software Engineer & Technical Lead, Platform Product *March 2022 – January 2023*

Software Engineer, Platform Product *April 2020 – February 2022*

- Developed domain-driven microservices to enable the automation of processing insurance claims and orders for patients' prescriptions. Performed foundational work with tools, frameworks, and architecture ideologies recommended by Business & Engineering leadership to assess their value for product vision.
- Facilitated ticket refinement and discovery sessions as Technical Lead of an Agile team with Product Managers, Product Owners, and Designers to advocate for technical product requirements and constraints.
- Designed, architected, and built a business-critical transaction processing (OLTP) system which supports over 30,000 transactions per-business day. Built dashboards and alerting for Engineering and Product Managers to monitor and observe the health of the system in Production.

- Led a multi-member project to refactor the top-of-funnel intake work stream for new users of the pharmacy. This workflow orchestrates the reception of new prescriptions and gathering information of new patients across all pharmacy locations nationwide. The new workflow follows an Event-driven Architecture and leverages the Local Read Model (LRM) pattern to deliver this data to distributed consumers. This workflow sees a peak throughput of over 500 messages per-second.
- Rewrote a proprietary EDI transmission library from the ground-up to enable product evolvability, improve tests and documentation, and retain intellectual control for long-term maintainability. Data integrity improved from 5.9% to 99.9%, using a sample size of nearly 400,000 historical transactions.

IBM, *New York, NY*

Software Developer & Technology Consultant, Blockchain Innovation Unit *February 2018 – April 2020*

- Engineered client solutions leveraging Linux Hyperledger Fabric blockchain technology.
- Managed and deployed a production blockchain network using Helm, Tiller, Docker, and Istio in a first-of-its-kind automated deployment of enterprise blockchain using Hyperledger Fabric on Kubernetes for a major financial services client.
- Committed Change Request [FAB-15051] to the official Linux Hyperledger Fabric open source project to fix a bug in their example "smart contract" code. Extended the blockchain application's use case to remove assets from a distributed ledger's world state (database).

Skills

Programming Languages: Python, Java, Kotlin, SQL, JavaScript, TypeScript

Tools: Spring Boot, GraphQL, Datadog, Redis, Valkey, Elasticsearch, ActiveMQ, Kubernetes, Docker, Django, Express, FastAPI, Flask, Helm, Node.js, Grafana, Loki, Prometheus, PromQL, LogQL, Linux Hyperledger Fabric, Git, Github, AWS Cloud, Google Cloud, KEDA, OpenTofu, Terraform, Github Actions

Methodologies: Agile Software Development, Test-driven Development, Domain-driven Design, Event Storming, Event-driven Architecture, Distributed systems, Microservices, Product & Project Management

Education

B.S. Computer Science, Rensselaer Polytechnic Institute, Troy, NY *August 2014 – December 2017*